

DATA STRUCTURE

List

1. List is defined using square brackets [].
2. `append()` adds a value to the end of the list.
3. Duplicate values are allowed in a list.
4. `len(list_name)` returns the number of elements in the list.
5. A list can hold multiple data types.
6. The `id()` function returns the unique identity (ID) of the list. Example: `id(l)`
7. `clear()` removes all elements from the list.
8. `copy()` creates a copy of the list. Example: `l2 = l.copy()`
9. `del()` deletes the list
10. `count()` function number of occurrences of value in list
11. `index()` Adds a value at a specific position.
12. `extend()` Adds multiple values from another list.
13. `remove()` Removes the first matching value.
14. `pop()` Removes and returns an element.
15. `pop(index)` Removes an element from a specific position
16. `index()` Returns the position of a value.
17. Slicing is allowed in list

```
In [2]: l = []  
l
```

```
Out[2]: []
```

```
In [3]: type(l)
```

```
Out[3]: list
```

```
In [4]: len(l)
```

```
Out[4]: 0
```

```
In [15]: l.clear()  
l.append(10)#only append one at a time  
l
```

```
Out[15]: [10]
```

```
In [16]: len(l)
```

```
Out[16]: 1
```

```
In [20]: l.append(10)
l.append(20)
l.append(30)
l.append(40)
l.append(50)
l.append(60)
l
```

```
Out[20]: [10, 10, 20, 30, 40, 50, 60]
```

```
In [21]: len(l)
```

```
Out[21]: 7
```

```
In [23]: l.append("adarash")
l.append(3.20)
l
```

```
Out[23]: [10, 10, 20, 30, 40, 50, 60, 'adarash', 3.2]
```

```
In [34]: l=[[1,2,3,4,5,6],10, 10, 20, 30, 40, 50, 60,'adarash', 3.2]
l
```

```
Out[34]: [[1, 2, 3, 4, 5, 6], 10, 10, 20, 30, 40, 50, 60, 'adarash', 3.2]
```

```
In [36]: l2=l.copy()
l2
```

```
Out[36]: [[1, 2, 3, 4, 5, 6], 10, 10, 20, 30, 40, 50, 60, 'adarash', 3.2]
```

```
In [38]: print('list1',l)
print('list2',l2)
```

```
list1 [[1, 2, 3, 4, 5, 6], 10, 10, 20, 30, 40, 50, 60, 'adarash', 3.2]
list2 [[1, 2, 3, 4, 5, 6], 10, 10, 20, 30, 40, 50, 60, 'adarash', 3.2]
```

```
In [39]: l==l2
```

```
Out[39]: True
```

```
In [40]: id(l)
```

```
Out[40]: 1856664308480
```

```
In [41]: id(l2)
```

```
Out[41]: 1856664272576
```

```
In [42]: id(l)==id(l2)
```

```
Out[42]: False
```

```
In [43]: id(l)!=id(l2)
```

```
Out[43]: True
```

```
In [46]: del(12)
12
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[46], line 1
----> 1 del(12)
      2 12

NameError: name '12' is not defined
```

```
In [47]: l.count(10)
```

```
Out[47]: 1
```

```
In [75]: l.insert(1, "Python")
1
```

```
Out[75]: [40, 'Python', 'Python', 4, 5, 1, 2, 3, 4]
```

```
In [58]: l2=[1,2,3,4,5,6]
```

```
In [59]: l.extend(l2)
1
```

```
Out[59]: [40, 'Python', 'Python', 30, 20, 15, 4, 5, 1, 2, 3, 4, 5, 6]
```

```
In [76]: l.remove(1)
1
```

```
Out[76]: [40, 'Python', 'Python', 4, 5, 2, 3, 4]
```

```
In [71]: l.pop()
```

```
Out[71]: 5
```

```
In [70]: l.pop(2)
```

```
Out[70]: 15
```

```
In [78]: print(l)
l.index(3)
```

```
[40, 'Python', 'Python', 4, 5, 2, 3, 4]
```

```
Out[78]: 6
```

```
In [82]: print(l)
l.index('Python')
```

```
[40, 'Python', 'Python', 4, 5, 2, 3, 4]
```

```
Out[82]: 1
```

```
In [83]: 1
```

```
Out[83]: [40, 'Python', 'Python', 4, 5, 2, 3, 4]
```

```
In [84]: l[:]
```

```
Out[84]: [40, 'Python', 'Python', 4, 5, 2, 3, 4]
```

```
In [85]: l[5]
```

```
Out[85]: 2
```

```
In [86]: l[9]
```

```
-----  
IndexError                                Traceback (most recent call last)  
Cell In[86], line 1  
----> 1 l[9]  
  
IndexError: list index out of range
```

```
In [89]: print(l)  
         l[:3]
```

```
[40, 'Python', 'Python', 4, 5, 2, 3, 4]
```

```
Out[89]: [40, 'Python', 'Python']
```

```
In [90]: print(l)  
         l[2:5]
```

```
[40, 'Python', 'Python', 4, 5, 2, 3, 4]
```

```
Out[90]: ['Python', 4, 5]
```

```
In [91]: l[:-3]
```

```
Out[91]: [40, 'Python', 'Python', 4, 5]
```

```
In [92]: l[::-5]
```

```
Out[92]: [4, 'Python']
```

```
In [100... print(l[1][0])  
           print(l[1][1])  
           print(l[1][2])  
           print(l[1][3])  
           print(l[1][4])  
           print(l[1][5])
```

```
P  
y  
t  
h  
o  
n
```